Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python/FastAPI** data pipelines, **PostgreSQL** backends, and **React/TypeScript** apps, operating production services with Docker, AWS, CI/CD, and Sentry observability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **ETL** pipelines with **FastAPI**, PostgreSQL, SQL Server, and cloud storage, running 5K+ monthly jobs with incremental loads, reconciliation checks, and error handling, sustaining 99%+ production reliability.
- Re-architected async LLM orchestration with FastAPI background workers, explicit job states, and **failure isolation**, processing 1K+ jobs and tripling throughput under concurrent load while query and index tuning cut dashboard latency 71%.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a full-stack **React** and Python platform deployed on **Render** and Vercel via GitHub Actions, serving 150+ instructors and 10K+ daily requests with stable latency during peak load.
- Designed a **concurrent batch-ingestion** pipeline with ThreadPoolExecutor for PDF, DOCX, and ZIP files, processing 1K+ files per batch with fault-tolerant execution and per-call token and cost logging.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **ETL** workflows with AWS Glue and S3, normalizing raw JSON, CSV, and Parquet data from multiple sources into analytics-ready formats, boosting pipeline efficiency 68% and cutting storage costs 15%.
- Optimized high-volume APIs with pagination, field filtering, indexing, and **caching**, improving response times 30%, and ran nightly Jenkins and GitHub CI pipelines with coverage reports across three operating systems.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight microservices with **React**, TypeScript, FastAPI, and PostgreSQL, streaming real-time WebSocket updates and synchronizing 2M+ records across locations for authentication, resource, billing, lead, and notification workflows.
- Instrumented production services with **Sentry**, Azure Monitor, and Kibana and shipped Docker builds through Jenkins CI/CD with unit, integration, and end-to-end tests, reaching 95%+ build validation across releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built hourly Python ingestion of 50K+ booking and payment events into **Snowflake** with **dbt** models, aggregating peak hours, service demand, daily active users, and revenue for internal dashboards.
- Developed **FastAPI** booking, pricing, and payment APIs with Firebase real-time sync and idempotent **webhooks**, sustaining sub-200ms latency and cutting manual refund handling 70% across multi-currency payment flows.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, Shell
Frontend: React, Next.js, Single-Page Applications, TailwindCSS, Shadcn, Data Visualization
Backend & Data: FastAPI, Flask, REST APIs, WebSockets, ETL Pipelines, dbt, Async Workers
Databases: PostgreSQL, Snowflake, ClickHouse, MySQL, MongoDB, Firebase
Cloud & DevOps: AWS (S3, Glue, RDS), Docker, Kubernetes, Render, Vercel, GitHub, GitHub Actions, Jenkins, CI/CD
Product & Reliability: Sentry, Kibana, Monitoring, Observability, Clerk, Stripe, Code Reviews, Performance Optimization, Technical Writing

Projects

Multi-Source Market Data Pipeline | *Python, FastAPI, ClickHouse, dbt, React, TypeScript, Docker, Sentry*

- Designed a scraper-driven ingestion pipeline normalizing public electricity-market feeds into **ClickHouse** via **dbt**, with retrying idempotent loads, freshness alerts in Sentry, a FastAPI query layer, and React time-series charts.

Big-Brain Live | *Clerk, Convex, PostgreSQL, FAISS, LLaMA-2, Shadcn*

- Built a document platform with **Clerk** authentication, Convex real-time collaboration, and a PostgreSQL-backed ingestion and indexing pipeline over 50K+ records with incremental updates, cutting retrieval latency 40%.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes, TailwindCSS*

- Built a **TypeScript** and Next.js fintech app with Plaid bank linking, WebSocket real-time updates, and advanced transaction filtering over 1M+ transactions, deployed on Kubernetes with CI/CD pipelines.